

AR(1) process

$$X_t = \alpha + \beta X_{t-1} + \varepsilon_t \quad \{\varepsilon_t\} \text{ white noise } WN(0, \sigma^2)$$

$$= \alpha + \beta [\alpha + \beta X_{t-2} + \varepsilon_{t-1}] + \varepsilon_t$$

$$= \alpha(1+\beta) + \beta^2 X_{t-2} + \varepsilon_t + \beta \varepsilon_{t-1}$$

$$= \alpha(1+\beta) + \beta^2 [\alpha + \beta X_{t-3} + \varepsilon_{t-2}] + \varepsilon_t + \beta \varepsilon_{t-1}$$

$$= \alpha(1+\beta+\beta^2) + \beta^3 X_{t-3} + \varepsilon_t + \beta \varepsilon_{t-1} + \beta^2 \varepsilon_{t-2}$$

$$\vdots$$

$$= \alpha \left(\sum_{i=0}^{t-1} \beta^i \right) + \beta^t X_0 + \sum_{i=0}^{t-1} \beta^i \varepsilon_{t-i}$$

$$\text{if } |\beta| < 1 \quad \alpha \sum_{i=0}^{t-1} \beta^i \rightarrow \frac{\alpha}{1-\beta} \quad , \quad \beta^t X_0 \rightarrow 0 \quad , \quad \sum_{i=0}^{t-1} \beta^i \varepsilon_{t-i} \rightarrow \sum_{i=0}^{\infty} \beta^i \varepsilon_{t-i}$$

if $|\beta| < 1$

$$X_t = \frac{\alpha}{1-\beta} + \sum_{i=0}^{\infty} \beta^i \varepsilon_{t-i}$$

MA(∞)

Wold representation of AR(1)

